EXP.NO: 5	Write a Servlet to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.
DATE:05/03/2025	

AIM:

To demonstrate the difference between the HTTP GET and POST methods by creating a form in an HTML page and handling the form submissions using two different HTTP methods in a Java Servlet.

ALGORITHM:

1. **Input:**

- The user fills out a web form with their name and email.
- The form will use both GET and POST methods for data submission.

2. **Steps for Creating and Handling the HTTP GET and POST Requests:**

1. **Create an HTML Form:**
    - Design an HTML form with two input fields: one for the name and one for the email.
    - Create two buttons to allow the user to submit the form using GET or POST methods.
    - Use the action attribute to define the Servlet URL and specify the form method (either GET or POST).
2. **Create the Servlet to Handle GET and POST Requests:**
    - Write a Java Servlet class that will handle both GET and POST requests.
    - The servlet will have two methods:
        - doGet() to handle the GET requests.
        - doPost() to handle the POST requests.

- In both methods, retrieve the form data using request.getParameter().
- In the doGet() method, process and display the data in the URL.
- In the doPost() method, process and display the data in the HTTP request body.

3. **Handle GET Requests in the Servlet:**
   - In the doGet() method, use request.getParameter() to retrieve the name and email submitted via GET.
   - Output the data as a response, including the name and email in the URL (visible to the user).
4. **Handle POST Requests in the Servlet:**
   - In the doPost() method, use request.getParameter() to retrieve the name and email submitted via POST.
   - Output the data as a response, displaying the name and email without exposing them in the URL (more secure).
5. **Return the Output:**
   - The Servlet should return an HTML page displaying the name and email entered by the user.
   - For the GET method, the output should be visible in the URL.
   - For the POST method, the output should be visible in the response body, not the URL.
6. **Deploy and Test the Servlet:**
   - Compile the Servlet class and deploy it to a Servlet container (like Apache Tomcat).
   - Test the form using GET and POST methods to see the difference in data submission.
7. **Observe and Compare GET and POST Results:**

- Test the GET method: The form data is sent in the URL.
- Test the POST method: The form data is sent in the HTTP request body, not visible in the URL.

PROGRAM:

Exp5.html:

```
<!DOCTYPE html>
<html>
<head>
  <title>Library System</title>
  <style>
    body {
      margin: 0;
      font-family: "Segoe UI";
      background: #f2fcff;
    }
    header {
      background: #004aad;
      color: white;
      padding: 15px;
      text-align: center;
      font-size: 22px;
    }
```

```
.container {
  width: 85%;
  margin: 30px auto;
  display: ffex;
  gap: 20px;
}
.card {
  flex: 1;
  background: white;
  padding: 20px;
  border-radius: 10px;
  box-shadow: 0 0 10px #ccc;
}
h3 {
  color: #004aad;
}
input, textarea {
  width: 100%;
  padding: 10px;
  margin: 8px 0;
}
button {
  width: 100%;
  padding: 10px;
  background: #004aad;
```

```
        color: white;
        border: none;
      }
    </style>
</head>
<body>
<header>Online Library System</header>
<div class="container">
    <div class="card">
      <h3>Search Books (GET)</h3>
      <form action="Servelt4" method="get">
        <input type="text" name="keyword" placeholder="Search by
keyword">
        <button type="submit">Search</button>
      </form>
    </div>
    <div class="card">
      <h3>Select a Book (POST)</h3>
      <form action="Servelt4" method="post">
        <input type="text" name="title" placeholder="Notice Title" required>
        <input type="text" name="dept" placeholder="Department" required>
        <textarea name="content" placeholder="Notice Content"
required></textarea>
        <button type="submit">Submit Book</button>
      </form>
    </div>
```

```
</div>
</body>
</html>
```

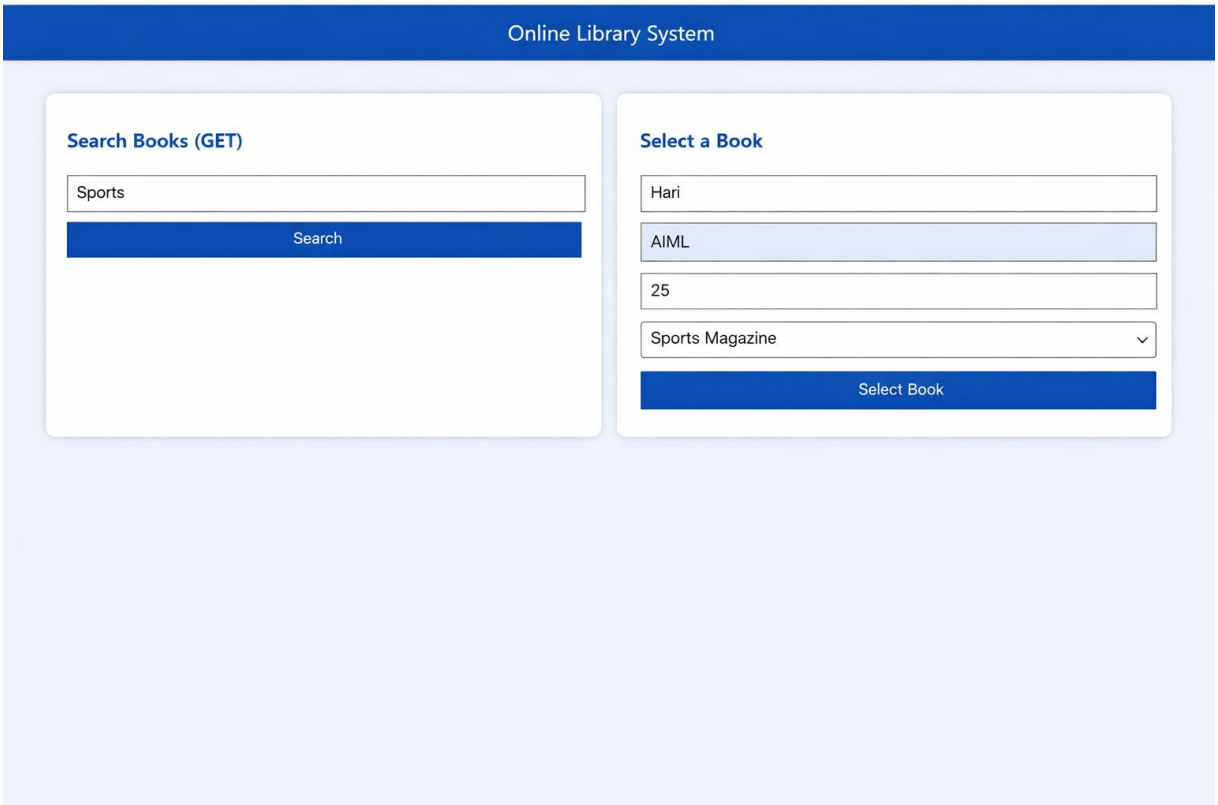
Servlet5.java:

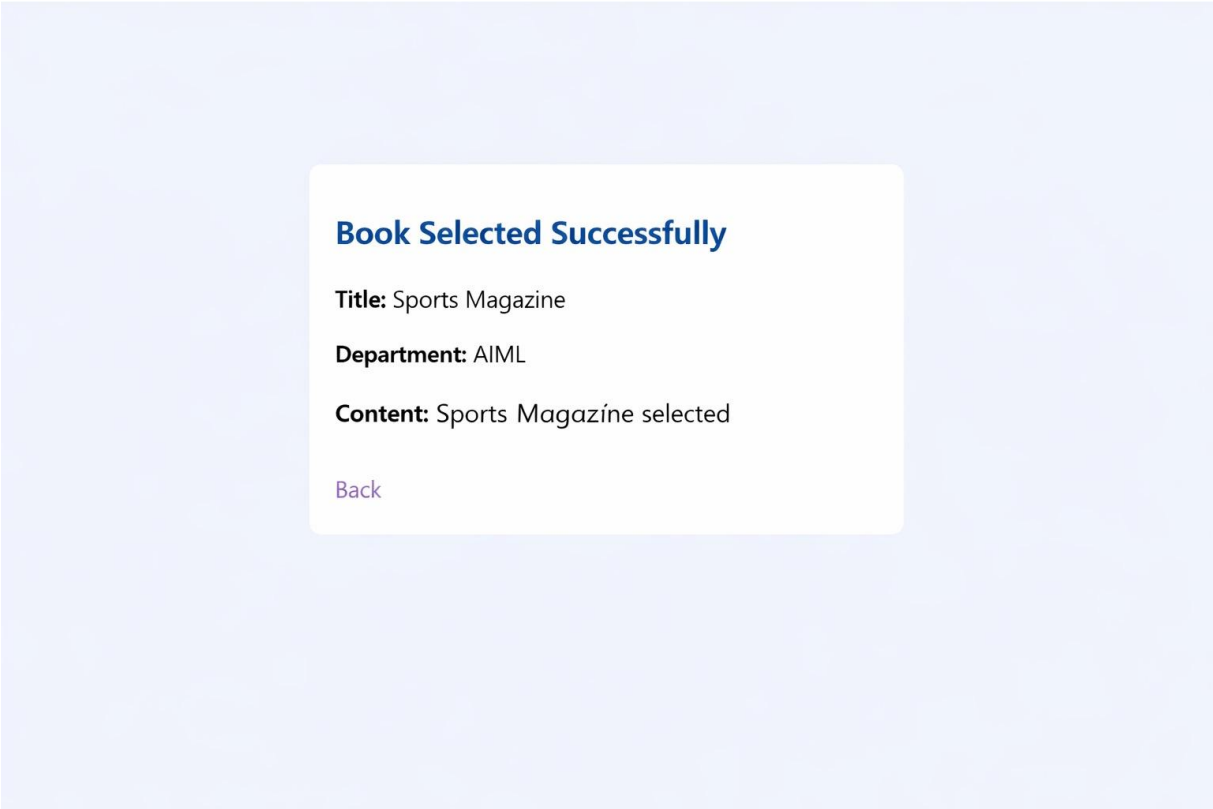
```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;
@WebServlet("/Servelt4")
public class Servelt4 extends HttpServlet {
   protected void doGet(HttpServletRequest request, HttpServletResponse
response)
         throws ServletException, IOException {
      response.setContentType("text/html");
      PrintWriter out = response.getWriter();
      String keyword = request.getParameter("keyword");
      out.println("<html><body style='font-family:Segoe
UI;background:#f2f6ff;'>");
      out.println("<div style='width:500px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");

      out.println("<h2 style='color:green;'>Search Results</h2>");
      if (keyword != null CC !keyword.isEmpty()) {
```

```
        out.println("<p>Showing results for: <b>" + keyword + "</b></p>");
      } else {
        out.println("<p>No keyword entered</p>");
      }
      out.println("<br><a href='Exp4.html'>Back</a>");
      out.println("</div></body></html>");
    }
    protected void doPost(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

      response.setContentType("text/html");
      PrintWriter out = response.getWriter();

      String title = request.getParameter("title");
      String dept = request.getParameter("dept");
      String content = request.getParameter("content");
      out.println("<html><body style='font-family:Segoe
UI;background:#f2f6ff;'>");
      out.println("<div style='width:500px;margin:50px
auto;background:white;padding:20px;border-radius:10px;'>");
      out.println("<h2 style='color:#004aad;'>Notice Posted
Successfully</h2>");
      out.println("<p><b>Title:</b> " + title + "</p>");
      out.println("<p><b>Department:</b> " + dept + "</p>");
      out.println("<p><b>Content:</b> " + content + "</p>");
```

```
        out.println("<br><a href='Exp4.html'>Back</a>");
        out.println("</div></body></html>");
    }
}
```

Output:





RESULT:

A Servlet is created to demonstrate the difference between HTTP GET and POST methods by creating a form and handling requests accordingly.